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10/2/25 Based on notes by: Jasmine Andresol, Kayden Kehe

Lecture 10: OT vs. Divergences

1 Review from Last Lecture

1.1 Gradient Flows

- **Motivation**: We often cannot compute an entire dynamic OT path or know the end result. Instead we can solve the problem gradually by minimizing function $\mathcal{F}(\rho)$.
- **Gradient flow**: As the step size of gradient descent approaches 0, we define a continuous curve starting at x_0 that follows ∇f
- **JKO scheme** (implicit Euler in probability space):

$$\rho_{t+1} = \arg\min_{\rho} \left\{ \mathcal{F}(\rho) + \tfrac{1}{2\tau} W_2^2(\rho, \rho_t) \right\}$$

- **PDE formulation**:

$$\partial_t \rho_t = \nabla \cdot \left(\rho_t \nabla \tfrac{\delta \mathcal{F}}{\delta \rho} \right)$$

- Recovers PDEs: heat, Fokker-Planck, etc.

1.2 Dynamic Formulation on Path Space

- Dynamic OT over space of paths:

$$W_2^2(\rho_0, \rho_1) = \min_{\bar{\pi} \in \Pi} \int_{\bar{X}} L(\gamma)^2 d\bar{\pi}(\gamma), \quad L(\gamma) = \int_0^1 |\dot{\gamma}(s)|^2 ds$$

- $L(\gamma) = \int_0^1 |\dot{\gamma}|^2 ds$ gives path length
- The optimal $\bar{\pi}^\star$ is equivalent to the Kantorovich solution (mass only on geodesics)
- Snapshots $\rho_t = (P_t)_\# \bar{\pi}^\star$

1.3 Schrödinger Bridges

- Dynamic formulation with diffuse paths. Amount of diffusion controlled by ϵ.
- Intuition: Consider stochastic paths that are not only geodesics.
- Problem:

$$\min_{\bar{\pi} \in \mathcal{U}(\rho_0, \rho_1)} \mathrm{KL}(\bar{\pi} \| \mathcal{K})$$

- Equivalent to entropic OT (solvable by Sinkhorn).

2 Today: OT vs Other Divergences

- Motivation & birds-eye view: section 3
- f-divergences: section 6
- Integral Probability Metrics (IPMs): section 7
- Trade-offs and comparisons: section 8

3 Motivation

- **Contextualization:** OT is one among many divergences.
- **Insight:** Contrasting with KL, MMD, Energy shows OT's special role.
- **Practice:** In practice, sometimes other divergences are more appropriate.
- This lecture will help us determine when OT is the right tool.

4 Three Ways to Compare Distributions

4.1 Horizontal (OT/Wassertein)

- Compare distributions globally by moving mass (broadly, OT is transportation / displacement).
- Sensitive to geometry: how far probability travels.
- Finite even if supports don't overlap.

4.2 Vertical (f-divergences, KL, JS)

- Compare locally via pointwise ratios $\alpha(x)/\beta(x)$ of densities.
- Sensitive to support overlap, infinity if supports don't match.
- Natural for likelihood/information-theory.

4.3 Functional(IPMS: MMD, etc)

- Compare via test functions f.
- Witness functions highlight distribution differences.
- Smoother than OT, more global than KL.
- Looking at distributions from the lens of a specific function class - function class imposes distance.

5 Formal Setup

- Let α and β be probability measures on $\mathcal{X}$.
- Divergence $D(\alpha\|\beta) : \mathcal{P}(\mathcal{X}) \times \mathcal{P}(\mathcal{X}) \to [0, \infty]$.
- Measures dissimilarity between probability distributions.
- Properties:
  1. Nonnegativity: $D(\alpha\|\beta) \geq 0$.
  2. Identity of indiscernibles $D(\alpha\|\beta) = 0 \iff \alpha = \beta$.
  3. Symmetry (not always satisfied) $D(\alpha\|\beta) = D(\beta\|\alpha)$.
  4. Triangle inequality (not always satisfied): $D(\alpha\|\beta) \leq D(\alpha\|\gamma) + D(\beta\|\gamma)$.

6 f-Divergences

6.1 Definition

Apply function to likelihood ratio $\alpha(x)/\beta(x)$. For convex entropy function f, f-divergence is:

$$D_f(\alpha\|\beta) = \int_{\mathcal{X}} f\left(\frac{d\alpha}{d\beta}(x)\right) d\beta(x)$$

Examples include KL, TV, Hellinger, χ^2, Jensen Shannon.
Entropy function f determines how to score discrepency ratio.

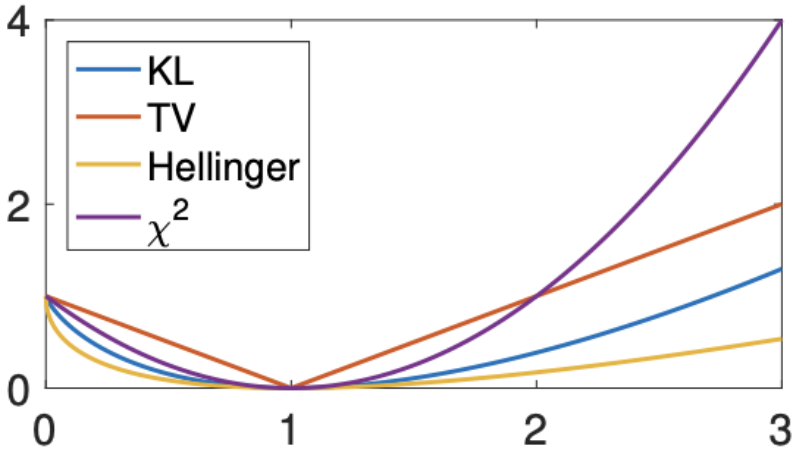


Figure 1: Entropy Functionals

6.2 Key Properties

- Nonnegativity, identity of indiscernibles. $D_f(\alpha\|\beta) \geq 0$, $D_f(\alpha\|\beta) = 0 \iff \alpha = \beta$.
- Asymmetric, no triangle inequality (not always a metric). $D_f(\alpha\|\beta) \neq D_f(\beta\|\alpha)$.
- Convexity, dual form via convex conjugates. Jointly convex in (α, β).
- Nice dual form: $D_f(\alpha\|\beta) = \sup_g\{\mathbb{E}_\alpha[g] - \mathbb{E}_\beta[f^\star(g)]\}$.

6.3 Use Cases in ML

- KL in variational inference (ELBO).
- JS in GANs.
- χ^2 in robust estimation.
- Regularization in RL (TRPO, PPO). Penalize divergence from prior or reference distribution,
- Can use $D_f(\alpha\|\beta)$ as a test-statistic in two-sample testing.
- Unsupervised learning (Hellinger in clustering, TV in imaging).

7 Integral Probability Metrics (IPMs)

7.1 Definition

With $\mathcal{F}$ as a function class, we find witness function f that gives worst-case expectation difference with α, β.

IPM is defined as follows:

$$d_{\mathcal{F}}(\alpha, \beta) = \sup_{f \in \mathcal{F}} \left(\mathbb{E}_\alpha[f] - \mathbb{E}_\beta[f]\right)$$

- $f : \mathcal{X} \to \mathbb{R}$ is called the witness function ('exposes' differences in probability).
- Choice of $\mathcal{F}$ class defines geometry/Form of the IPM.

7.2 Intution

$$d_{\mathcal{F}} = \sup_{f \in \mathcal{F}} \int_{\mathcal{X}} f(x) d\alpha(x) - \int_{\mathcal{X}} f(x) d\beta(x)$$

- IPMs quantify distance through a witness that exposes differences in probability

- They seek the best witness that maximally amplifies differences.
- If $d\alpha(x) >> d\beta(x)$, f makes x **large** and **positive**.
- If $d\alpha(x) << d\beta(x)$, f makes x **large** and **negative**.
- If $d\alpha(x) \approx d\beta(x)$, f can do whatever.

7.3 Maximum Mean Discrepancy (MMD)

MMD is an IPM, gives distance between mean embeddings of the distributions.

$$\text{MMD}_k(\alpha, \beta) = \|\mathbb{E}_{x\sim\alpha}[k(x, \cdot)] - \mathbb{E}_{y\sim\beta}[k(y, \cdot)]\|_{\mathcal{H}_k}$$

- **Intuition:** Embeds each distribution into a feature space defined by K.
- **Properties**:
  - $MMD_k(\alpha, \beta) = 0 \iff \alpha = \beta$.
  - No optimization required.
  - Often estimated by sample averages, computationally efficient.

7.4 Energy Distance

$$\|\mu - \nu\|_{ED} = \left(\mathbb{E}\|X - Y\| - \tfrac{1}{2}\mathbb{E}\|X - X'\| - \tfrac{1}{2}\mathbb{E}\|Y - Y'\|\right)^{1/2}$$

- This is just an MMD with kernel $k_{ED}(x, y) = -d(x, y)^p$ for $0 < p < 2$.
- Removes bandwidth tuning required for typical MMD.
- Popular in classical stats.

7.5 IPMs on Diracs

- W_1 is linear.
- Flat is like W_1 but it saturates at 1.
- Energy and Gaussian distance (both MMDs) show curvature.

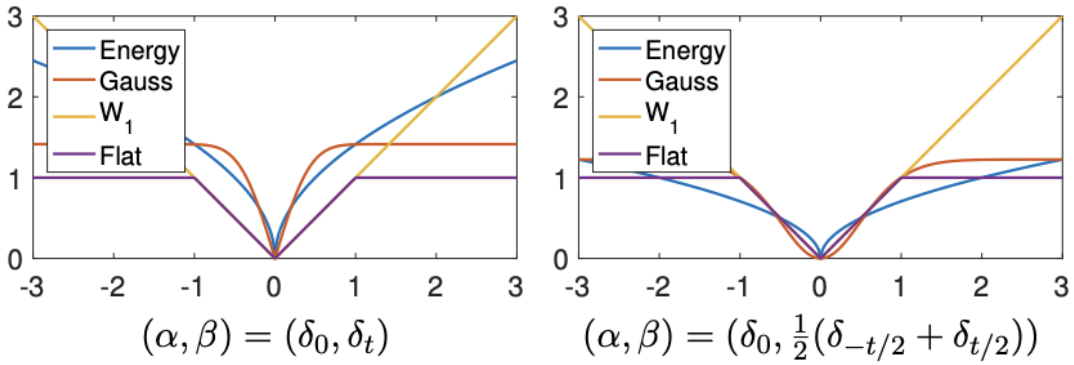


Figure 2: Comparison of dual norms

7.6 Witness Function Comparison

- Compares best witness function found by MMD and W_1.
- W_1 is piecewise linear, more flexible.
- Gaussian / MMD is smoother but less flexible.
- Spectrum of complexity (from least to most restrictive $\mathcal{F}$):
  TV $\to$ W_1 $\to$ Sobolev $\to$ MMD $\to$ Energy $\to$ KS / Test
- General Tradeoff
  - Richer $\mathcal{F}$ $\to$ stronger notion of closeness, more geometry, harder estimation, increase in expressiveness, decrease in statistical efficiency,
  - Simpler $\mathcal{F}$ $\to$ easier estimation, weaker geometry, less sensitivity, decrease in expressiveness, increase in statistical efficiency,

7.7 IPM Properties

- Metric Like
  - Always ≥ 0, equals $0 \iff \alpha = \beta$.
  - Symmetry: $d_{\mathcal{F}}(\alpha, \beta) = d_{\mathcal{F}}(\beta, \alpha)$.
  - Triangle inequality holds if $\mathcal{F}$ is convex and symmetric.
- Weak convergence
  - With smooth $\mathcal{F}$, IPMs metrize weak convergence (like OT, unlike TV or KL).
  - Examples: Wasserstein, MMD, Energy Distance.
- Witness function $f^\star$
  - Exposes where α and $beta$ differ.
  - Defined only up to sign, sometimes up to additive constants.

7.8 Discrepency Taxonomy

- W_1 is an IPM and an OT.
- W_p for $p > 1$ is OT, but not an IPM.
- Energy distance, KS, Sobolev, and MMD are IPMs.
- TV distance is an IPM and an f-divergence.
- Hellinger, JS, KL, and χ^2 are all f-divergences.

8 Comparisons and Trade-offs

8.1 KL, TV, JS vs OT

- OT: metric, reflects geometry, finite on disjoint supports.
- KL: asymmetric, ∞ if supports disjoint.
- TV: metric, ignores geometry.
- JS: bounded, softer than KL.

	KL	TV	JS	Wasserstein
Symmetry	✗	✓	✓	✓
Finite if supports disjoint?	✗ (∞ if no overlap)	✓	✓	✓
Metric?	✗	✓	✗ (square root is a metric)	✓
Sensitivity	Very strong (blows up)	"All-or-nothing" differences	Softer, bounded in [0,\log 2]	Smooth, reflects geometry
Interpretation	Relative entropy (information loss)	Max probability mismatch	Average divergence to midpoint distribution	Optimal transport cost
ML usage	Variational inference, ELBO	Hypothesis testing	GAN loss (original)	WGAN, domain adaptation, barycenters

Figure 3: Distance Comparisons

8.2 Interpolation: OT, Sinkhorn, MMD

- Trade-off between computational/statistical complexity, geometry.
- Coupling structures: hard (OT), soft (Sinkhorn), naive (MMD).

8.3 Geometry Awareness

- OT induces Euclidean geometry (e.g. Gaussian case). Straight line geodesics.
- KL undefined if supports don't overlap. Curved arc geodesics. Induces hyperbolic geometry.
- OT interpolates *linearly* in (m, σ) space with Gaussians. KL interpolates non-linearly.
- TV saturates and gives the maximum possible deviation.

8.4 Hilbertian Metrics

- **Definition:** A metric d on set $\mathcal{X}$ is Hilbertian if there exists an embedding $\Phi : \mathcal{X} \to \mathcal{H}$ such that $d(x, y) = \|\Phi(x) - \Phi(y)\|_{\mathcal{H}}$.
- **Intuition:** The geometry of d is just Euclidean hidden in a possibly (∞-dim) feature space.
- **Importance:** Hilbertian metrics admit positive definite kernels. Many efficient algorithms (kernel methods, random projections, convex opt) rely on this.
- **Examples:** MMD, Energy, Hellinger, and χ^2 are Hilbertian. W_p $(d \geq 2)$, KL, JS, TV, and most f-divergences are non-Hilbertian.

9 When to Use What?

- **OT:**
  - Yes, if geometry of support matters (no obvious way to use OT when geometry is unclear).
  - Yes, if supports might not overlap.
  - No, if in high dimensions where computational cost becomes intractable.
- **f-divergences:**
  - Yes, when densities overlap (histograms) and likelihood ratios are meaningful.
  - Yes, if you want to connect it to information theory.
  - No, if supports are disjoint (divergence becomes ∞).

- **IPMs:**
  - Yes, when efficient, dimension-free estimation is needed.
  - Flexible via choice of function class $\mathcal{F}$.
  - No, when geometry heavily depends on the function class.

10 Takeaways & Preview

- OT, f-divergences, and IPMs are three major families of statistical divergences.
- Trade-off between geometry, smoothness, tractability.
- Next time: OT as ML loss (differentiability, mini-batching, backprop).